

Sparsen LASSO Induced by Convolutional Expansion (SLICE) for Robust Peak Localization in Gamma Spectroscopy

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Abstract

Gamma-ray spectroscopy often requires precise feature extraction directly in the time domain to handle pile-up events. While sparse regression techniques like LASSO are natural candidates for this task, they suffer from a "peak splitting" phenomenon: even when the dictionary is perfectly known, the solver fractures single physical events into clusters of coefficients. We introduce SLICE (Sparsen LASSO Induced by Convolutional Expansion). By convolving the dictionary with a spreading kernel, we enforce structural sparsity. We demonstrate, through simulations and experimental data, that this approach, although counterintuitive, yields more robust pulse localization.

Keywords: Sparse Regression, Gamma Spectroscopy, Peak Splitting, LASSO

1. Introduction

Nuclear spectroscopy is a fundamental tool for identifying radioactive isotopes in fields ranging from nuclear security to astrophysics [1]. In most applications, emitted particles are collected by a detector, which converts their energies into electrical pulses. The primary objective is to measure both activity and contents of the unknown source, based on these recorded pulses. In

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7 high-count-rate scenarios, "pile-up" occurs; pulses overlap in time, distorting
8 the amplitude measurements and resulting energy histogram.

9 Recent efforts were made to address this problem in the time domain, e.g.,
10 using deep learning techniques [2, 3, 4]. However, the lack of annotated datasets
11 significantly limits the application of fully supervised methods in the field. This
12 motivates the use of classical ML methods, such as sparse regression techniques
13 as investigated by the authors in previous contributions [5, 6]. In this latter
14 framework, the problem can be modeled as $\mathbf{y} = \mathbf{A}^{(0)}\boldsymbol{\theta} + \boldsymbol{\varepsilon}$, where $\mathbf{y}$ is the digi-
15 tized detector output, $\mathbf{A}^{(0)}$ is a convolutional dictionary of shifted pulse shapes,
16 $\boldsymbol{\varepsilon}$ is an additive noise, and $\boldsymbol{\theta}$ represents the sparse stream of photon arrival
17 amplitudes. In [7, 5], the authors used a post-processed version of the Least
18 Absolute Shrinkage and Selection Operator (LASSO) to extract the individ-
19 ual pulses localization, which provided satisfactory results for low-to-medium
20 counting rates. However, a critical issue arises in this context, which is later
21 denoted by peak splitting. Even for synthetic signals where the convolutional
22 dictionary $\mathbf{A}^{(0)}$ is completely known, standard LASSO fails to recover the cor-
23 rect support [8]. Due to the high mutual coherence between columns of $\mathbf{A}^{(0)}$,
24 the optimization approximates a single physical peak using a cluster of adjacent
25 columns to fit noise or grid mismatches. This issue has also been noted in sparse
26 coding and convolutional dictionary learning settings, for which regularization
27 and further optimization techniques have been suggested [9, 10].

28 This paper builds on previous contributions by the authors and tries to
29 solve the peak splitting issue inherent to sparse regressors with convolutional
30 dictionaries. To address this, we introduce SLICE (Sparser LASSO Induced
31 by Convolutional Expansion). The model introduced in subsection 2.2 adds a
32 blurring, convolutional matrix which low-pass filters the regressor $\mathbf{x}$ plugged in
33 entry. Equivalently, we compute the LASSO with a linearly transformed version
34 of $\mathbf{A}^{(0)}$ (4), whose columns are blurred versions of the original ones.

35 Though this might appear counterintuitive at first (as after such an opera-
36 tion the mutual coherence is expected to have increased), it might be understood
37 as adding some structural robustness: since the LASSO with $\mathbf{A}^{(0)}$ may be prone

to peak splitting, in contrast SLICE reparameterization has the effect of activating simultaneously several atoms of the original dictionary $\mathbf{A}^{(0)}$ pertaining to neighboring sampling times, which can yield a sparser regressor fulfilling the main task of explaining the bulk of the signal, as seen in Figure 1. The approach

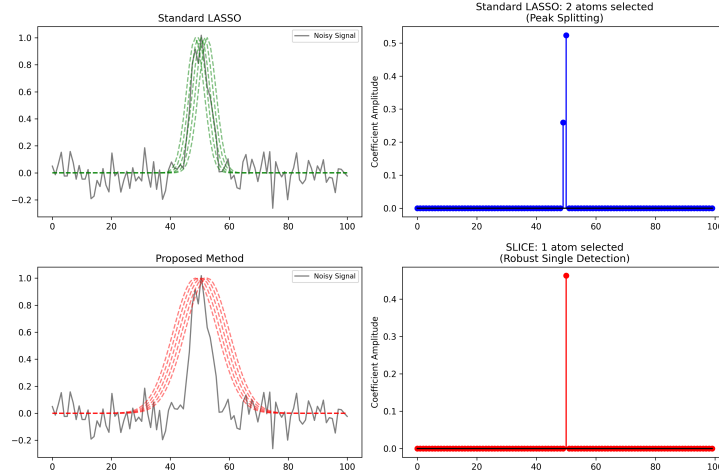


Figure 1: Illustration of the impact of applying positive convolutions to a basic shape for a synthetic signal. The top row shows that even when provided the actual shape (upper left), LASSO splits the peaks in an attempt to improve accuracy (upper right). In contrast, SLICE smooths the regressors, forcing the optimizer to be robust and pick a single column that best matches the bulk of the signal.

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42 sacrifices reconstruction resolution for improved localization, which is crucial in
 43 the field. The rest of the paper provides a description of the algorithm and
 44 results on synthetic data and real signals for activity estimation.

45 2. Overview of Sparser LASSO Induced by Convolutional Expansion 46 (SLICE)

47 2.1. Model and Assumptions

48 We assume that the signal observed is of the form of a discretized shot-noise
 49 process, that is

$$y[n] = \sum_{k \geq 1} h_k(t_n - T_k) + \varepsilon[n], \quad n = 0 \dots N - 1 \quad (1)$$

where in (1) $\{T_k, k \geq 1\}$ is the sample path of a Poisson process, $\varepsilon[n]$ denotes a sequence of independent and identically distributed Gaussian random variables modeling the noise, and h_k denotes the shape of at the k^{th} -th electrical pulse generated by an impinging particle, assuming to have a Gamma shape, that is $h_k(t) = t^\alpha e^{-\kappa t} 1(t > 0)$. Therefore, we build an initial dictionary of p shapes following this assumption with parameters $(\alpha_k, \kappa_k) \in [0.4, 1.3]^2, k = 1 \dots p$ (all fully known). Assuming the sampling frequency is large so that discretization errors can be neglected, (1) can be rewritten as

$$\mathbf{y} = \mathbf{A}^{(0)} \boldsymbol{\theta} + \boldsymbol{\varepsilon}, \quad (2)$$

where $\mathbf{y} := [y[0] \ y[1] \ \dots \ y[N-1]]^T$, $\boldsymbol{\varepsilon} = [\varepsilon[0] \ \varepsilon[1] \ \dots \ \varepsilon[N-1]]^T$, $\boldsymbol{\theta}$ is a sparse vector whose entries are positive if the entry index corresponds to an occurrence of the underlying process, and equal 0 otherwise, and $\mathbf{A}^{(0)} \in \mathbb{M}_{N \times pM}$ is a convolutional dictionary made of translated versions of h_k as $\mathbf{A}_{i, np+k}^{(0)} = h_k[i-1-n]$. Since we are interested in peaks localization, our aim is to estimate the following support in time: $S(\boldsymbol{\theta}) := \{m \mid \|\boldsymbol{\theta}|_{B_m}\|_1 > 0\} \subseteq \{0 \dots M-1\}$ where $B_m := \{mp+1, \dots mp+p\}$.

2.2. Description of SLICE

As previously discussed, SLICE relies on a reparameterized variable (equivalently, on a transformed version of the original dictionary). Given a nonnegative, ℓ_1 -normalized discrete-time window Λ with support $\{-q, \dots, q\}$ and its induced convolution matrix $\mathbf{K}$ (applied shape-wise), the SLICE problem becomes:

$$\hat{\mathbf{x}}_{\text{SLICE}} = \arg \min_{\mathbf{x} \geq 0} \left\{ \frac{1}{2} \|\mathbf{y} - (\mathbf{A}^{(0)} \mathbf{K}) \mathbf{x}\|_2^2 + r \|\mathbf{x}\|_1 \right\}, \quad (3)$$

where r is the standard LASSO sparsity parameter fixing a trade-off between precision and sparsity. Hence a SLICE regressor is a LASSO with positivity constraint, computed for the modified dictionary:

$$\mathbf{A} = \mathbf{A}^{(0)} \mathbf{K} \quad (4)$$

Interestingly, since the columns of $\mathbf{K}$ are ℓ_1 -normalized (excepting those pertaining to the first β and last β sampling times), (3) can be interpreted as

essentially solving the LASSO for $\mathbf{A}^{(0)}$ on the proper subcone of $\mathbb{R}_+^M$ defined by $\mathcal{C}_{\mathbf{K}} = \mathbf{K} \mathbb{R}_+^M$ (proper as soon as $\mathbf{K} \neq \mathbf{I}_M$). When Λ is the impulse response of a low-pass filter, as in our applications, such a transformation typically spreads out the atoms supports. Furthermore, since $\mathbf{A}$ is also a convolutional dictionary, the proximal gradient method can also be substantially accelerated [11, 12] for practical implementations. Note also that though we focus on a natural block partition of the regressor entries, this approach quite differs from the Group LASSO [13], which employs an $\ell_{2,1}$ penalty (i.e. modifies the penalty term) and does not attenuate the peak splitting effect along neighboring groups which is our core issue.

The peaks' localizations can be estimated afterwards from the thresholded block-support $S(\hat{\mathbf{x}}; \eta) := \{i \in [M] \mid \|\hat{\mathbf{x}}_{B_i}\|_1 > \eta\}$ [7]. Though one might compare directly $S(\hat{\mathbf{x}}_{\text{SLICE}}; \eta)$ to $S(\boldsymbol{\theta})$, it is impractical as errors by a few sampling points in the block-sparsity pattern occur but are not critical. Instead, sparsity recovery performance shall be quantified as follows: given a positive integer β , and for each $m \in S(\boldsymbol{\theta})$ and the β -neighborhood $V_\beta[m] = \{m - \beta, \dots, m + \beta\}$, we define the concentration measure as

$$\mathcal{C}(\mathbf{x}) = \frac{\|\mathcal{P}_{\mathcal{V}_\beta} \mathbf{x}\|_1}{\|\mathbf{x}\|_1}, \quad (5)$$

where $\mathcal{P}_{\mathcal{V}_\beta}$ denotes the projection onto the subset of coefficients $\mathcal{V}_\beta := \bigcup_{m \in S(\boldsymbol{\theta})} V_\beta[m]$.

(5) quantifies the proportion of recovered peak energy falling within a tolerance neighborhood of the true support. It rewards robust detection instead of exact support recovery while penalizing the peak splitting artifacts. Though it depends on a user-defined radius β , it does not depend on any threshold-based pruning of active sampling times as in [7]. Results presented in section 3 below show that provided Λ is adequately chosen, SLICE active coefficients tend indeed to be more concentrated around the points of $S(\boldsymbol{\theta})$ than the other methods tested.

101 3. Results

102 In this section, we present results on peak localization using SLICE obtained
 103 on simulations and real data. We investigated in our simulations two specific
 104 cases:

- 105 (I) The shapes are drawn randomly from $\mathbf{A}^{(0)}$; though this case is not realistic
 106 in practice, it allows a direct comparison with $S(\boldsymbol{\theta})$, and
- 107 (II) Each electrical pulse is defined by (α, κ) randomly drawn in the interval
 108 $[0.4, 1.3]^2$; this allows to assess the robustness of the proposed approach in
 109 the case of a moderate mismatch between the true shapes and those used
 110 for reconstruction.

111 We set $\lambda \in \{0.05, 0.1, 0.15, 0.2, 0.3, 0.4\}$, to investigate performance when
 112 pile-ups can be encountered with an increasing probability. For each activity,
 113 100 sample paths of (1) with $N = 500$ sampling points are generated; these
 114 sample paths give the true sparsity pattern. The energies $\{E_n, n \geq 0\}$ are drawn
 115 accordingly to the Gaussian mixture $0.8\mathcal{N}(100, 10) + 0.2\mathcal{N}(225, 15)$ truncated
 116 at 0, and the noise variance is chosen $\sigma = 1$, in accordance with Signal-to-Noise
 117 Ratios commonly found in Gamma spectrometry applications. The number of
 118 shapes was set to $p = 4$, and we investigated the influence of the convolution
 119 kernel's variance by using the normalized kernels family $\Lambda(\sigma) \propto G_K$, where G_K
 120 is a fixed truncated Gaussian shape with standard deviation $\sigma_K \in [0.05; 3]$.
 121 The results obtained with SLICE are compared with results from LASSO, the
 122 Elastic Net [14], SCAD [15], and the ℓ_1/ℓ_2 quotient minimization introduced in
 123 [16]. We choose $r = 10$ as it yields the best results for LASSO in terms and
 124 activity estimation and reconstruction. The additional parameter in the Elastic
 125 Net and SCAD are set to 1 and 3.7, respectively.

126 3.1. Simulations Results and Discussion

127 Results obtained on the concentration for 4 shapes with 90% confidence
 128 intervals are displayed in figures 2 for case (I) and 3 for case (II), for a chosen

neighborhood radius $\beta = 3$ and $\lambda = 0.15$. As expected, when σ_K is small LASSO and SLICE behave similarly. We can furthermore observe that the proposed approach outperforms the other compared algorithms, though the Elastic Net and the close-to-optimal LASSO also perform quite well. In both cases, there exists a significantly wide interval of σ_K values such that SLICE performs better in terms of the concentration measure (5).

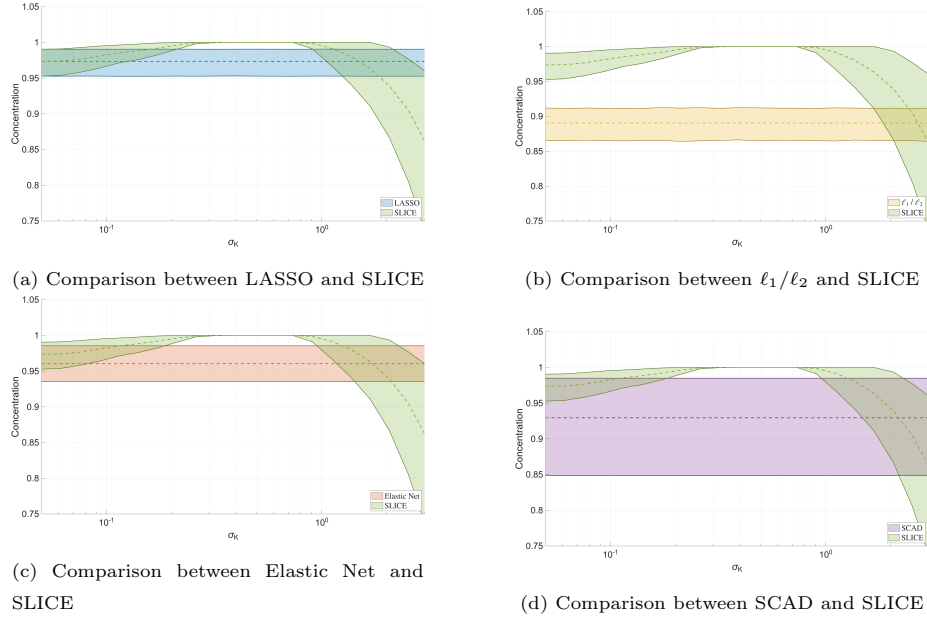


Figure 2: Evolution of the concentration as a function of the kernel bandwidth σ_K in $\mathbf{K}$ and Comparison with different algorithms with 90% confidence intervals - case (I).

Note that even in case (I), SLICE can yield sparser results in terms of concentration for an adequate choice of σ_K ; when σ_K is too high, SLICE performs poorly due to the dictionary inability to reconstruct $\mathbf{y}$. To better illustrate the connection between precision in peak localization and precision in peak reconstruction for $\lambda = 0.15$, Figure 4 displays the relationship between the Root Mean Squared Error (RMSE) between $\mathbf{y}$ and its reconstruction and the concentration for SLICE applied to 100 simulated signals. We considered three levels of dictionary smoothing, $\sigma_K \in \{0.01, 0.5, 1\}$, corresponding to almost unsmoothed,

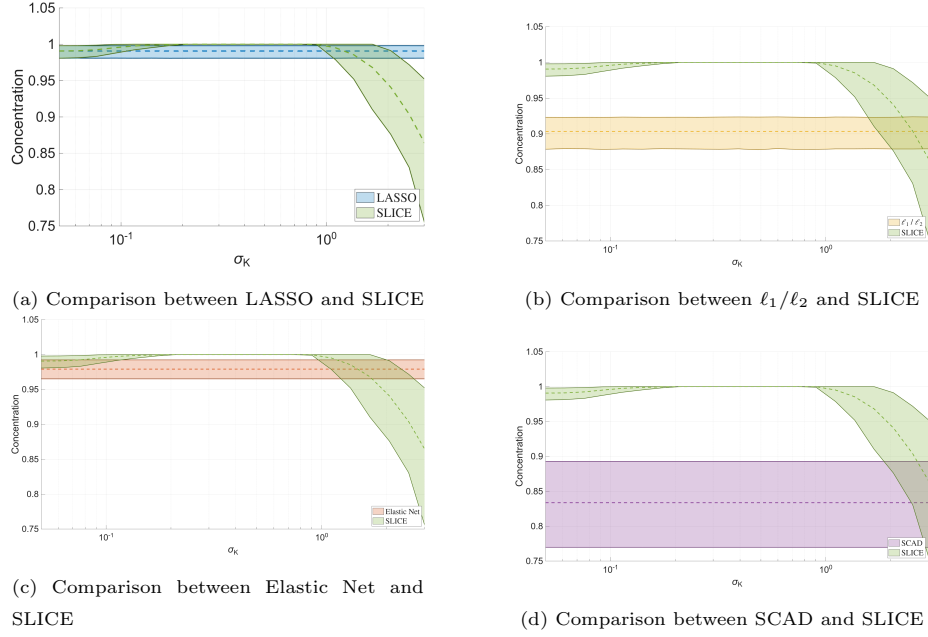


Figure 3: Evolution of the concentration as a function of the kernel bandwidth σ_K in $\mathbf{K}$ and comparison with different algorithms with 90% confidence intervals - case (II).

moderately smoothed, and over-smoothed dictionaries, respectively. The first

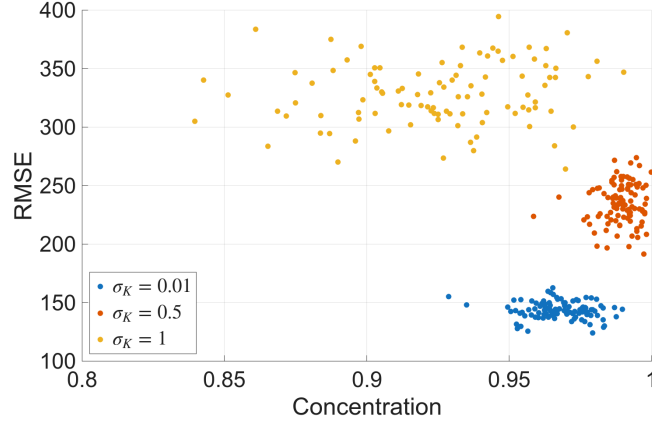


Figure 4: RMSE vs Concentration for three SLICE smoothing levels - experiments were performed for 100 signals in case (I), $\lambda = 0.15$ and 4 shapes in $\mathbf{A}^{(0)}$.

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144 regime behaves similarly to applying LASSO with $\mathbf{A}^{(0)}$ and, as expected, yields

	$\mathcal{C}(\hat{\mathbf{x}}_{\text{SLICE}})$	Precision	Recall	F1
Rectangular	0.98	0.45	0.99	0.62
Triangle	0.99	0.44	0.99	0.62
Epanechnikov	0.99	0.45	0.99	0.62
Gaussian	0.98	0.42	0.99	0.60
Right-side Gaussian	0.64	0.43	0.99	0.60

Table 1: Performance metrics (average values obtained for 100 tests) obtained for different kernels for Case (I). Experiments performed on Case (II) provided closest to identical results.

the lowest RMSE values because the regressors are closely matched to the dictionary that generated the signals. In contrast, the third regime produces a widely scattered RMSE–concentration plot with substantially higher RMSE and concentration, reflecting the strong mismatch and introduced by excessive smoothing. Interestingly, the intermediate smoothing level achieves noticeably higher concentration scores overall, albeit with a moderate increase in RMSE. Results displayed indicate the existence of a good interval for σ_K , if one wishes to sacrifice a bit of precision in the reconstruction to allow better localization. Inference on a sensible choice for the kernel bandwidth remains to be investigated in future contributions.

Table 1 shows that the kernel choice has little influence of the results obtained for $\lambda = 0.15$ with a bandwidth $\sigma_K = 0.5$, provided it is even; bounded support of the kernel might have a slightly positive influence on the results obtained. Similar behavior was observed for other activities.

In order to characterize peak localization performance, we compare four metrics: Precision and Recall, Average Block Ratio and Concentration. All measures (besides $\mathcal{C}(\cdot)$, which does not rely on thresholding) rely on a fixed threshold $\eta = 10$. As is known, for convolutional dictionaries, all sparse reconstruction methods perform poorly in terms of occurrences recovery in the hard sense after thresholding. However, in activity estimation, slight errors in peak localization can be overlooked as long as the correct number of peaks is

166 determined. Thus, we adapt the definition true positive and false positives at a
 167 neighborhood level, as now presented.

168 Given a radius β and a tested regressor $\hat{\mathbf{x}}$, the recall is defined as

$$\text{Rec}_\delta(\hat{\mathbf{x}}) := \frac{\#\{m \in S(\boldsymbol{\theta}) ; S(\hat{\mathbf{x}}; \eta) \cap V_\beta[m] \neq \emptyset\}}{\#S(\boldsymbol{\theta})}, \quad (6)$$

169 that is, a true positive is associate to m as long as $V_\beta[m]$ includes one active
 170 block of $\hat{\mathbf{x}}$. For each $V_\beta[m], m \in S(\boldsymbol{\theta})$, the index in $S(\hat{\mathbf{x}}; \eta) \cap V_\beta[m]$ associated
 171 with the block with the largest ℓ_1 norm is referred by as true positive, while the
 172 others are marked as false positive. With this update, the original definition of
 173 the precision can be computed as a function of β . Note that $\beta = 0$ gives the
 174 original precision and recall. In addition, we compute the Average Block Ratio
 175 (ABR) of a given regressor thresholded $\hat{\mathbf{x}}$, defined as

$$\text{ABR}(\hat{\mathbf{x}}) = \frac{\#S(\hat{\mathbf{x}}; \eta)}{\#S(\boldsymbol{\theta})}, \quad (7)$$

176 as a global sparsity indicator. Table 2 summarizes the results obtained for all the
 177 metrics for two neighborhood radii $\beta \in \{0, 2\}$, for all the tested activities. Recall
 178 reaches near-perfect values for all methods when $\beta = 2$, indicating that each
 179 algorithm activates sampling times sufficiently close to the true occurrences.
 180 When $\beta = 0$, recall drops substantially across all methods, which supports the
 181 use of our softer, neighborhood-based definition in (6) rather than the stricter,
 182 traditional criterion that requires $m \in S(\hat{\mathbf{x}}; \eta)$ for every $m \in S(\boldsymbol{\theta})$. For mild
 183 activities $\lambda \in [0.05 \ 0.2]$, all methods exhibit higher precision for $\beta = 2$ than
 184 for $\beta = 0$. In this regime, true events are typically well separated, so slightly
 185 larger neighborhoods capture more occurrences and increase the number of false
 186 positives.

187 Overall, SLICE dominates precision scores, with SCAD coming close to or
 188 slightly outperforming it. Elastic Net provides rather balanced results, while
 189 ℓ_1/ℓ_2 provides good recall, which indicates better localization capability. ABR,
 190 as a global indicator, shows that SLICE provides an overall sparser solution.
 191 However, since this criterion depends on η and does not provide local insights,
 192 it does not reward estimates that select blocks closer to the true events, which

can be a limitation when accurate peak localization is required. On the other hand, $\mathcal{C}(\cdot)$ depends solely on the fraction of the ℓ_1 norm contained in the neighborhood $\mathcal{V}_\beta$ around $S(\boldsymbol{\theta})$, regardless of how this mass is distributed across $\mathcal{V}_\beta$, which is better for the soft peak localization we aim to. Conversely, $\mathcal{C}(\cdot)$ strongly penalizes regressors whose ℓ_1 norm is scattered across small contributions located remotely from the true occurrences. This shows some superiority of the concentration as a threshold-independent metric. As illustrated in Table 2, in the $\lambda = 0.15$ row, for instance, SLICE and SCAD exhibit markedly different concentration values despite having similar and comparatively favorable ABR scores.

For $\beta = 2$ and for a broad range of activities $\lambda \in [0.05 \ 0.3]$ (and at a lesser level for $\lambda = 0.4$), concentration provides a clearer discriminative power than precision, separating SLICE, LASSO and Elastic Net from SCAD and ℓ_1/ℓ_2 .

The influence of the additive noise level on the concentration of all methods was further investigated. The average concentration as a function of the noise standard deviation for $\beta = 2$, $\sigma_K = 0.5$ and 100 simulated signals (for case (I)) is displayed on Figure 5. The SLICE shows the greatest resilience to noise level increase, whereas LASSO and Elastic Net perform quite similarly. This is not surprising, since highly correlated design features are not a decisive factor for concentration scores.

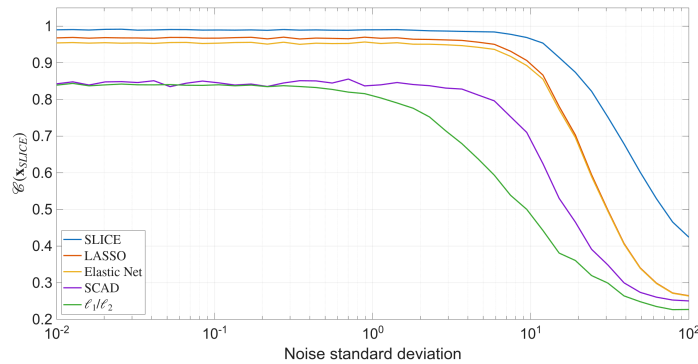


Figure 5: Concentration vs noise level for the different methods

213 3.2. Results and Discussion on Real Spectrometric Data

214 We tested SLICE and LASSO on a real γ -spectrometry signal used in [7] for
 215 peak localization. The original dictionary $\mathbf{A}^{(0)}$, the sparsity parameters, and
 216 the transformation $\mathbf{K}$ were chosen similarly to the ones used in the simulation.
 217 A direct comparison of the recorded signal and the block-sparsity patterns is
 218 shown in figure 6.

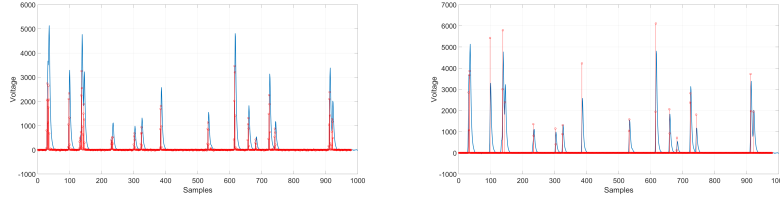


Figure 6: Block sparsity patterns for LASSO (left) and SLICE (right), compared with the original signal

219 One of the common issues in the field is that most recordings are not anno-
 220 tated, thus preventing the comparison to a ground truth. We thus illustrate the
 221 performance of SLICE numerically based on the average number of active sam-
 222 pling times and the estimated source activity. Part of the real signal was split
 223 into 100 segments of 500 samples; on each segment, LASSO and SLICE with
 224 $\sigma_K = 3$ were applied both with $r = 10$. In both cases, a dictionary of 4 shapes
 225 was applied, whose parameters were taken as $(\alpha, \kappa) \in \{0.3 + 0.1k, k = 0 \dots 3\}^2$
 226 (these values were used in [7, 5] to get a preliminary plausible fit with the elec-
 227 trical pulses). Given a threshold value, for each obtained regressor, we compute
 228 two criteria presented in Figure 7, both as functions of the threshold: (a) the av-
 229 erage number of active blocks (across the 100 segments), and (b) the estimated
 230 source activity derived from the number of active times (as in [7]). Generally
 231 speaking, the previous figure shows that SLICE is less sensitive to threshold
 232 variations than LASSO. Also, it selects fewer blocks overall. In both cases, the
 233 estimated activity is scaled by the number of connected components in the ac-
 234 tive pattern, and, in that respect, SLICE is also more stable, with a significantly
 235 reduced overall bias.

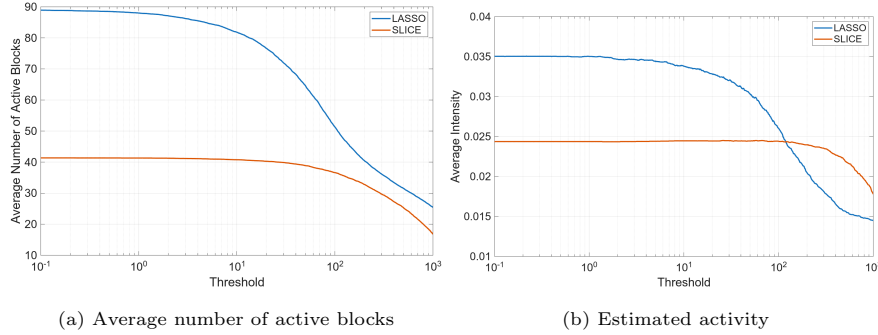


Figure 7: Results obtained on real data for LASSO (blue) and SLICE (red). The actual activity is $\lambda_{true} = 0.02$.

236 The comparison between the two methods using actual data shows that
 237 SLICE outperforms the original LASSO in terms of sparsity patterns, peak lo-
 238 calization capabilities and activity estimation. This point is crucial for practical
 239 applications, as overlapping pulses starting at close arrival times can easily be
 240 misidentified as a single occurrence. For example, we can see from the block-
 241 sparsity patterns between sampling times 0 and 200 that close pile-ups of pulses
 242 are better separated using SLICE rather than the LASSO with $\mathbf{A}^{(0)}$. This point
 243 is very encouraging for real-life applications since, in nuclear spectroscopic appli-
 244 cations, the observed signals can be extremely long (up to hundreds of millions
 245 of sampling points). Combined with a fast acceleration scheme such as the one
 246 developed in [11, 12], the use of SLICE can yield fast processing based on lighter
 247 dictionaries.

248 4. Conclusion

249 This paper introduces SLICE, a modification of LASSO that provides sparser
 250 solutions, and demonstrates, through simulations and real data, that it can
 251 be a suitable alternative to other sparse reconstruction methods when precise
 252 localization is more important than reconstruction accuracy. Recoverability,
 253 changes in cross-correlation structures, or error bounds are valuable theoretical
 254 insights into this approach, and should be investigated in future contributions.

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Table 2: Performance Metrics across different methods and activities

λ	Method	$\mathcal{C}(\hat{\mathbf{x}})$		Precision		Recall		ABR
		$\beta = 0$	$\beta = 2$	$\beta = 0$	$\beta = 2$	$\beta = 0$	$\beta = 2$	
0.05	SLICE	0.46	0.99	0.34	0.48	0.58	0.99	1.12
	LASSO	0.46	0.96	0.33	0.47	0.65	0.99	1.30
	Elastic Net	0.45	0.94	0.33	0.46	0.68	0.99	1.36
	SCAD	0.41	0.82	0.30	0.44	0.52	0.99	1.20
	ℓ_1/ℓ_2	0.40	0.76	0.32	0.44	0.73	0.99	1.52
0.1	SLICE	0.55	0.99	0.34	0.46	0.58	0.99	1.10
	LASSO	0.54	0.96	0.33	0.42	0.64	0.99	1.26
	Elastic Net	0.54	0.95	0.34	0.43	0.68	0.99	1.35
	SCAD	0.49	0.86	0.32	0.44	0.51	0.99	1.10
	ℓ_1/ℓ_2	0.49	0.81	0.33	0.42	0.72	0.99	1.49
0.15	SLICE	0.51	0.99	0.36	0.41	0.61	0.99	1.09
	LASSO	0.49	0.97	0.35	0.42	0.68	0.99	1.27
	Elastic Net	0.49	0.95	0.35	0.41	0.72	0.99	1.36
	SCAD	0.45	0.85	0.32	0.42	0.53	0.99	1.11
	ℓ_1/ℓ_2	0.45	0.85	0.33	0.39	0.74	0.99	1.49
0.2	SLICE	0.53	0.98	0.36	0.43	0.60	0.99	1.07
	LASSO	0.52	0.97	0.35	0.41	0.69	0.99	1.26
	Elastic Net	0.52	0.96	0.35	0.40	0.73	0.99	1.36
	SCAD	0.47	0.75	0.32	0.40	0.54	0.99	1.13
	ℓ_1/ℓ_2	0.48	0.86	0.33	0.37	0.75	0.99	1.51
0.3	SLICE	0.57	0.99	0.38	0.41	0.62	0.99	1.04
	LASSO	0.55	0.97	0.37	0.38	0.72	0.99	1.23
	Elastic Net	0.54	0.97	0.37	0.37	0.76	0.99	1.31
	SCAD	0.50	0.79	0.33	0.38	0.57	0.99	1.13
	ℓ_1/ℓ_2	0.51	0.87	0.35	0.35	0.78	0.99	1.46
0.4	SLICE	0.61	0.99	0.39	0.39	0.64	0.99	1.01
	LASSO	0.59	0.98	0.38	0.36	0.73	0.99	1.21
	Elastic Net	0.58	0.98	0.37	0.35	0.76	0.99	1.27
	SCAD	0.54	0.91	0.35	0.34	0.59	0.99	1.14
	ℓ_1/ℓ_2	0.56	0.91	0.36	0.32	0.79	0.99	1.43